Music-specific challenges. Compared to domains such as books, movies, or e-commerce [56, 124, 193], music consumption presents a number of particular challenges for user modeling in the context of MRS:

(1) Music catalogs are extremely large, often spanning millions of tracks. Moreover, music catalogs tend to be very rich in terms of culture, music genres, year of production etc. As a result, users interact with hundreds or even thousands of unique tracks over time.
(2) On music streaming platforms, the most common form of feedback is implicit. Implicit feedback refers to user actions and behaviors that are not explicitly provided as signals of preference. In this setting, feedback is typically derived from listening time or from binary listening events defined by a time threshold, often set at 30 seconds. The challenge stems from implicit feedback being a noisier signal of interest than explicit ratings or reviews [147]. Moreover, in the absence of explicit textual feedback, user preference modeling must rely on item metadata, whose quality varies considerably [100].
(3) User musical preferences are complex, often spanning both short- and long-term dimensions [144]. Short-term preferences reflect recent explorations and can be quite noisy, while long-term preferences capture more stable tastes developed over years. However, long-term preferences may also become outdated and lack novelty. Moreover, musical preferences are highly contextual. For example, soundtracking—selecting music for a given activity, is a very common way of consuming music [50, 98]. In different contexts, users will change the type of music they consume to match the mood or the energy pertaining to their current activity: someone will play energetic music when working out or calm music for studying. This can drastically change the user's preferences, something that a static user representation might not be able to pick up.
(4) Affective responses can be quite subjective in nature when it comes to music. So are their impact on music preferences. An additional factor making the use of affective cues challenging is that music is a highly multimodal artifact [33] and different channels of delivery (e.g., audio, lyrics, album cover, video clip) may elicit different emotions. Moreover, emotions may dynamically evolve over the time a user listens to a music piece. How to acquire and integrate such emotional trajectories in the recommendation process is still an intricate problem.

Key changes introduced by LLMs. With the introduction of LLMs, a new alternative to traditional opaque user embeddings has become viable: the use of natural language representations of user preferences [56, 124, 132, 144, 193]. In this setting, LLMs act as summarization devices, producing high-level representations of users' preferences from collections of interacted items. Based on these generated profiles, recommendations can be produced in two main ways: (i) by employing another LLM as a ranker to score candidate items given the user profile, or (ii) by embedding both user profiles and item metadata into a shared latent space, enabling a more traditional recommendation scheme. The use of textual taste representations offers many advantages, particularly their interpretability. Because these profiles are understandable, users can see what the system "knows" about them, increasing transparency and building trust. Moreover, in cold-start situations, users could explicitly state their preferences, enabling the system to generate recommendations. Furthermore, if users are allowed to edit their profiles, they could take control of their recommendations by steering them toward new interests and away from content they no longer find relevant.

A common method for deriving natural language preferences with LLMs is to concatenate explicit feedback, such as item reviews, and metadata from consumed items into prompts and use the LLM to summarize this information into a profile [56, 124, 144, 193]. It is widely acknowledged that such profiles need to be concise in order to be scrutable. In this process, LLMs aggregate the most relevant information encoded in users' consumption histories. In a second step, after profiles are generated or collected, recommendation can again proceed in multiple ways: either by projecting both

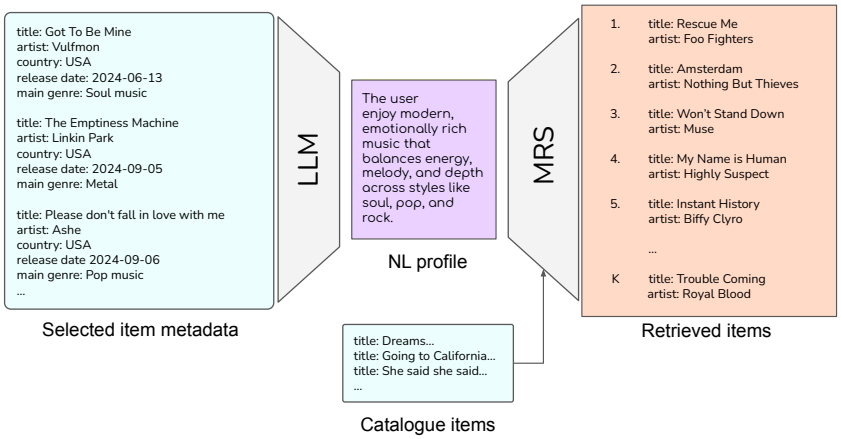

Fig. 2. Representation of the method to derive NL user preference profiles from consumption data.

user profiles and item metadata into a shared latent space [56, 144] or by employing a second LLM, employed here as recommender (described in Section 2.3). A representation of this method is shown in Fig. 2.

Given that LLMs are nowadays often perceived to exhibit human-like responses and demonstrate traces of "thinking" or reasoning, it is natural to investigate — and potentially leverage — the *psychological traits* they may simulate, adapting concepts and mechanisms from human psychology such as personality, emotions, or cognitive effects and biases. LLMs could enable new ways of approaching personality-related tasks in the domain of music recommendation. For instance, they can be used to create personality profiles of their users (e.g., by predicting traits from the user's prompting history or by carrying out a conversation with the user guided towards inferring their personality using relevant personality assessment questionnaires) [152, 184]. Enriched with correlation models between personality and music traits or preferences such as those proposed in [95, 103], LLMs could then further tailor recommendations accordingly. Furthermore, LLMs appear to infer persona-related aspects such as musical diversity, popularity, or mainstreaminess [9, 83] from the user's listening history with some success. Research on persona-based music recommendation can draw from recent large datasets of LLM-generated personas, e.g. [59], that provide a synthetic dataset of 1 billion personas. While currently not tailored to the music domain, corresponding research can inform the creation of music-specific persona datasets, which in turn could be used to (1) support the classification of users into personas from their conversation with the LLM and (2) facilitate the simulation of music-related conversations with the system.

Regarding listening intents, LLMs provide a great tool to consolidate and extend these findings from music psychology. For instance, integrating a data-driven approach with the results of the empirical questionnaire-based study from [140], Hausberger et al. leverage LLMs to augment and cluster 129 listening functions into 32 broader intents and provide a simple interface that enables music search by intent [62]. Another pathway could be to infer the listening intent of the users from their conversations with an LLM, either through additional classifiers or directly through LLM-guided questions. Ultimately, the listening intent could be used by traditional recommendation approaches, integrating it as contextual factor, or again directly by the LLM. Surprisingly little literature has been devoted to using LLMs for finding music that fits a specific *emotion*. Nevertheless, LLMs have shown some potential to identify emotions from conversations [49, 54]. Linking these to individual music preferences and using the resulting user model to tailor music

recommendations, however, is still an open challenge with only very recent and limited research available [162, 164]. This is a highly valuable research direction given the importance of mood and emotion in music perception and use.

2.2 Item Modeling

While collaborative filtering recommender systems primarily rely on user–item and item co-occurrence in streaming sessions, content-based or hybrid MRS focus on a deeper understanding of an *item's features* during the recommendation process. These features align better with how humans perceive music, offering more support for explainable recommendations, enabling discovery, and item cold-start situations [33, 138].

Background. Item features relevant to the MRS range from low-level attributes extracted from audio or lyrics to higher-level features such as user-generated content (e.g., playlist titles or music tags). Deldjoo et al. [33] propose an "onion model" to categorize musical content relevant for recommendation, placing *audio* at the core, followed by *embedded metadata* from music providers (e.g., title, artist, album, lyrics, label, recording year), and then outer layers of increasingly expressive, subjective, or culturally influenced features that reflect personal or community perceptions of music. In particular, *expert-generated metadata* often focuses on musical aspects such as genre, style, or mood, but may also include details about music properties (e.g., instrumentation, vocal attributes, lyrical themes) or artist information like biographies. *User-generated metadata* includes item descriptions such as tags, lyric interpretations, and music reviews, typically created on social media or forums (e.g., last.fm[1], reddit[2], SongMeanings[3]) or within music streaming platforms when naming playlists or interacting with tracks after a search.

Embedded metadata and editorial information, such as genres, are typically available when music is ingested into streaming services. However, this data could be incomplete, ambiguous or inaccurate. To tackle this, past works have explored matching musical items with knowledge bases or editorial encyclopedias, as well as aligning existing musical taxonomies or graphs to improve metadata verification, completeness, and music contextualization (e.g., adjusting music tagging taxonomy to a specific country) [41, 42].

Another approach to describing music—its moods, styles, and properties such as instrumentation, has been to rely directly on the audio signal. Many works in the music information retrieval (MIR) community focus on tasks such as audio auto-tagging [70] or automatic audio classification [123] (e.g., by genre, mood, or listening context); voice and instrument identification [126]; lyrics transcription [195], when these are unavailable; and music captioning [53, 58], which has significantly improved with the advancement of language generation models.

When richer textual data related to music, such as reviews or music/artist descriptions, is available, established natural language processing (NLP) techniques could be used for aspect extraction [11], named entity recognition [40], and sentiment or negation identification [130], as detailed in Table 1.

Music-specific challenges. Unlike domains such as book, news, or e-commerce recommendations, music content is largely non-textual apart from lyrics. However, lyrics are copyrighted, making their use difficult in a setting of reproducible research, as experimental datasets cannot be shared.

In the context of music streaming services, textual information is often limited to short forms, and common NLP techniques struggle with these short or noisy texts [3] (e.g. user tags or playlist titles, transcribed user queries). When richer textual data is available, such as music reviews or social media posts, the text nature of track titles and artist

[1] https://www.last.fm
[2] https://www.reddit.com/r/rock/
[3] https://songmeanings.com

Extractive Task	Description
Tag Extraction	Extract attributes or tags of music items from text (e.g., music genres, lyrical themes, instruments or voice characteristics)
Named Entity Recognition (NER) *Coreference Resolution*	Detect names or mentions of music items (e.g. artists, albums, or tracks) from text. In creative domains like music, challenges stem from the frequent use of common words to denote named entities, and input text being often very noisy (e.g., user natural language queries or transcribed voice commands).
Sentiment or Negation Identification	Identify music items that are discussed with negative sentiment or explicitly disliked, which can be valuable for personalization and content filtering.

Table 1. Extractive tasks for data augmentation when richer descriptions of music items are available.

names introduces additional challenges [40]. Their vocabulary often overlaps with tagging terms. For example, "love" might refer to a track title, a mood, or a lyrical theme. This ambiguity complicates key tasks such as named entity recognition and artist disambiguation (linking to the catalog or knowledge base). Moreover, detecting negations as a way of expressing dislike (e.g., disliked music artists or genres) remains a significant challenge, even for LLMs [57, 156].

When relying on audio as input, music auto-tagging was until recently limited to fixed vocabularies, performing well for some tags but failing with contextualized or user-specific ones (e.g., music for "travel", "running", or "relaxing") [70]. Similarly, early audio captioning works, prior to transformer-based models, often produced unnatural or even grammatically flawed descriptions [38]. Although newer models have improved linguistic fluency [53, 58], studies show that audio is not always effectively leveraged as a modality [169].

Finally, music, as a cultural product, is perceived differently across communities[17, 43]. The same artist may receive different genre tags depending on the localization of Wikipedia editors or the annotation practices of music providers or listeners. This variability is even greater for mood- or context-related tags, where personalization is essential but often absent (e.g., "party music" can mean very different things to different users [102]). Consequently, aggregating and reconciling diverse knowledge sources for tag annotation, or music description generation in general, remains a significant challenge.

Key changes introduced by LLMs. LLMs provide powerful capabilities for augmenting item information and have been leveraged in several complementary roles, each addressing different data-augmentation needs, as follows:

LLM as Annotator or Knowledge Base. Trained on vast amounts of online text, LLMs have become alternatives to knowledge bases or human annotators, even in zero-shot settings [6, 154, 186]. Thus, LLMs could generate user-friendly tags or keywords that describe a track, including its music genre or style. LLMs could also be used to suggest editorial metadata such as release dates, country of origin, or artist details, when these are missing. Fine-tuned LLMs specialized in the music domain can go further by producing rich, descriptive content about items [39]. Compared to other knowledge bases, LLMs can offer a degree of personalization in generating music metadata or descriptions, whether through prompt manipulation (e.g., including characteristics of the target user or community), in RAG-based solutions that interface with multiple tag taxonomies, or when used as personalized AI agents [25, 97].

LLM as Information Extractor. LLMs have proven highly effective at extracting specific information from unstructured text without even requiring a domain-adaptation or fine-tuning phase [61, 134]. For example, given album